

First Name:

Last Name: Answers

School of Computer Science
CS 2614: Computer Organization
Spring 2025
Quiz -2

Q. 1 Consider the following numbers represented as **unsigned** 5-bit binary numbers. Indicate whether the addition operation will generate any overflow.

01101 + 00110

$$\begin{array}{r} \overset{1}{0} \overset{1}{1} 1 0 1 \quad (13) \\ + 0 0 1 1 0 \quad (+6) \\ \hline 1 0 0 1 1 \quad (19) \checkmark \end{array}$$

No overflow

Q. 2 Consider the following numbers represented as **signed** 5-bit binary numbers. Indicate whether the addition operation will generate any overflow.

01101 + 00110

$$\begin{array}{r} \overset{1}{0} \overset{1}{1} 1 0 1 \quad (13) \\ + 0 0 1 1 0 \quad (+6) \\ \hline 1 0 0 1 1 \quad (-13) \times \\ \downarrow \text{2's complement} \end{array}$$

01101

Overflow

$$C_{in} = 1$$

$$1 \oplus 0$$

$$C_{out} = 0$$

1 $\rightarrow$ overflow